Is there a research-practice dosage gap in aphasia rehabilitation?

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Introduction

Motivation

- Few studies have evaluated the translation and implementation of evidence-based aphasia interventions to clinical practice[1]
- Dosage may be challenging to translate from research to clinical practice settings due to pragmatic clinical barriers such as insurance or transportation
- Anecdotes suggest limited dosage in practice but few comparisons to research
- A dosage mismatch between research and clinical practice threatens the external validity of treatment research & risks attenuating intervention effectiveness.

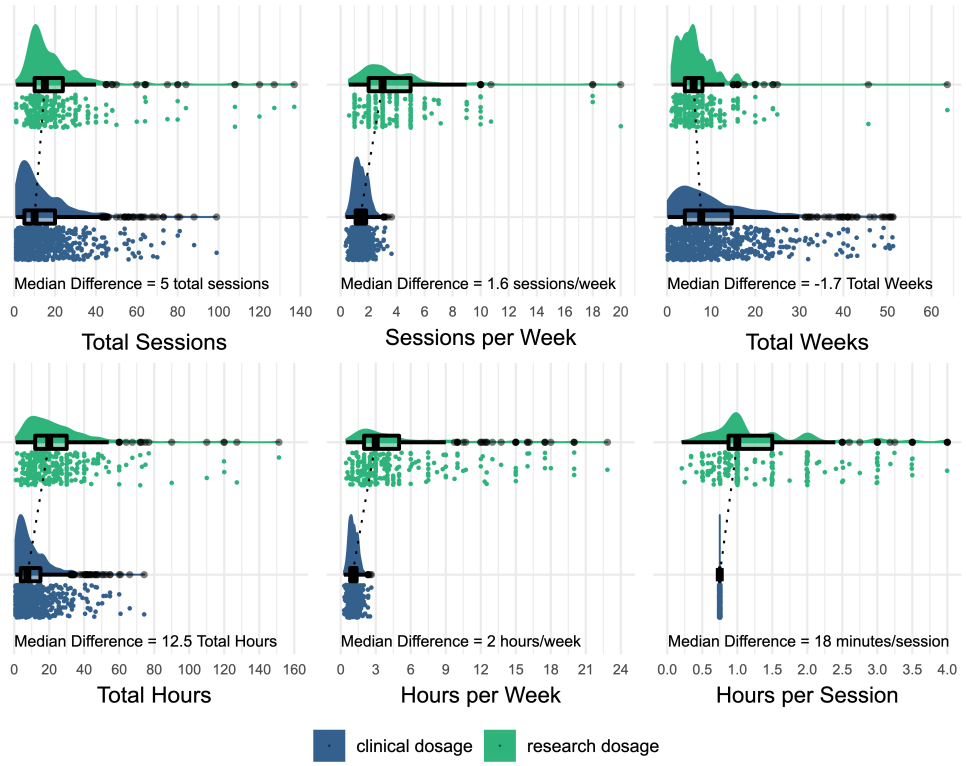
Research Questions

1. What is the typical treatment dose in an episode of care in outpatient clinical practice?

2. What is the typical treatment dose administered in contemporary clinical aphasia research?

3. To what extent is the dosage in treatment research aligned with clinical practice?

Differences in dosage between aphasia treatment research and clinical practice



Discussion

Take-aways

- Clinical dosage is frequently less than typical research dosage, except for total treatment duration.
- Total treatment hours and weekly intensity is particularly disparate between settings.
- Comparison does not include differences in priorities between research & clinical settings. Research settings focus on (a single) treatment program, clinicians incorporate multiple approaches, counseling, education, content with paperwork.
- Home practice appears more common in clinical practice,[2] could reduce disparity in dosage.
- Importance of treatment dosage does not supersede treatment efficacy and therapeutic value

Recommendations

1. Select dosage thoughtfully in clinical research, considering clinical practice constraints.
2. More research establishing dose-form & evaluating effects of different dosages.[3]
3. Take steps to facilitate home practice for higher-dose interventions (apps and low-tech materials.[4])
4. Pragmatic trials to evaluate implementation of aphasia treatments in clinical practice.
5. More advocacy needed for access to services, including intensive, comprehensive programs.

Methods

Clinical treatment dosage estimated via billing data from the Centers for Rehabilitation Services (CRS), primarily in Western Pennsylvania.

- Inclusion: Patients receiving an evaluation (CPT: 92523, 96105) and treatment (92507) from an SLP from 2014-2019 with ICD 9/10 diagnoses of aphasia and stroke.

Research dosage estimated via scoping review of prospective, aphasia treatment studies from 2009-2019

- Extracted dosage variables and estimated additional non-reported variables if possible
- Reliability (% agreement) for article inclusion/exclusion and data extraction >90%

Analysis: Non-parametric permutation tests[5]

Results

	Clinical Dosage				Research Dosage			
Variable	Mean	Median	Min	Max	Mean	Median	Min	Max
Total sessions*	14.50	10.00	5.00	20.00	20.10	15.00	10.00	23.80
Total hours*	10.90	7.50	3.80	15.00	25.10	20.00	12.00	30.00
Hours/session*	0.75	0.75	0.75	0.75	1.30	1.00	0.90	1.50
Hours/week*	1.10	1.10	0.80	1.40	4.70	3.00	2.00	5.00
Sessions/week*	1.50	1.40	1.10	1.80	3.60	3.00	2.00	5.00
Total weeks*	10.60	7.70	4.00	14.60	7.00	6.00	4.00	8.00

*Notes: 683 episodes of care included in CRS dataset. Standard treatment session is 45 minutes. Frequency calculated for episodes of care with >= 4 sessions. 303 Treatment studies included in review. > 82.5% reported sufficient details to calculate all dosage variables. *denotes difference in medians: p < 0.001"*

References & Acknowledgements

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